No. **5a**

Revised 1930

MAINTENANCE REGULATION

STANDARD PRACTICE FOR REPAIRS TO LOCOMOTIVE AND LOCOMOTIVE TYPE BOILERS

CANADIAN NATIONAL RAILWAYS

MOTIVE POWER DEPARTMENT

MONTREAL

5. METHOD OF SETTING SMALL TUBES IN BACK TUBE SHEET:

Copper ferrules for back tube sheet must be small enough to enter tube hole, and apply by placing same on style "A" prosser expander and tightening by driving pin twice in each ferrule with small pneumatic chipping hammer Drive ferrules in flush with sheet on fire side and allow surplus copper to project into water side *Use only No 14* B W G copper ferrule five-eighths of an inch wide for one-half inch tube sheet three-quarters of an inch wide for five-eighths inch tube sheet. When placing tubes in boiler particular care should be taken not to injure or displace the copper ferrules Ends of tubes must extend through *sheets five-sixteenths of an inch and must not be clinched* with hammer Style "B' prosser must then be used taking particular precaution that end of tube is close up against shoulder of prosser expander, changing prosser expander three times in each tube, so that tubes will be expanded uniformly The expander must be *of proper size corresponding* to the *thickness of* tube sheet, and long stroke pneumatic hammer must be used for this work, S 50 gun to be used for roundhouse work and S 60 gun at back shops.

After tubes have been thoroughly prossered, no further work must be done until after hydrostatic test has been completed, in order that the test may determine *whether tubes have been properly prossered* Tubes will then be beaded care being taken when beading to hold the tool so as to give the bead an outward set and to get the bead down to the sheet without raising a burr on the inside, or marking the sheet on the outside Belling or lapping of tubes before beading *is* not allowed

For tools to be used, reference must be made to the latest issues of the following drawings:—

Method of applying 2 & 2¼" tubes	3H-15174
" " 3½" "	3H-10725
2" Sectional Prosser styles A B C	3H-11269
2¼" " " D E F	3H 9546
3½" " " H J K.	3H 10724
Beading Tools	3H-15245

6. METHOD OF SETTING SMALL TUBES IN FRONT TUBE SHEET:

Tubes must extend through the front tube sheet not less than one-eighth inch and not more than 3/8" The only tubes to be beaded in the front end are *those shown on the tube sheet Drawings*

After tubes have been prossered in the back tube sheet they are to be rolled in the front tube sheet.

Any approved manufacturer s roller expander may be used for this operation

Shimming front tubes *is* not a *recommended practice*